

SRIRAM DEVATA

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Education

University of Illinois at Urbana-Champaign (UIUC) <i>PhD in Computer Science, Advised by Prof. Sarita Adve</i>	Aug. 2024 – Present 3.83/4.0
International Institute of Information Technology, Hyderabad (IIITH) <i>Bachelor of Technology (Honours) in Computer Science and MS by Research in Computational Natural Sciences</i>	Aug. 2019 – July 2024 8.99/10

Experience

University of Illinois at Urbana-Champaign (UIUC) <i>Graduate Research Assistant</i> - My broad research focus is to build software and hardware for datacenters. - Currently working on a serving system for compound AI workloads for datacenter GPUs. Preliminary results show that this serving system decreases the amount of GPU resources needed by 2x.	Aug. 2024 - Present
STARLab, Carnegie Mellon University <i>Student Researcher</i> - Worked with Prof. Akshitha Sriraman and Dr. Gilles Pokam to identify and leverage hardware design opportunities to enhance efficiency in datacenters by optimizing redundant datacenter “tax” operations. - Developed a methodology that uses perf to profile widely-used open-source microservices to uncover patterns that can guide the development of general hardware accelerators.	Jan. 2022 - June 2024
University of Waterloo <i>MITACS Globalink Research Intern</i> - Developed an innovative scoring metric in collaboration with Dr. Aravindhan Ganesan for evaluating and ranking ligand molecules designed by AI-driven generative models. - Tailored for the SARS-CoV-2 Main Protease’s primary binding site, this metric integrates interaction fingerprints and subpocket occupancy maps to offer a holistic approach to assessing molecular efficacy.	May 2023 - Aug. 2023
CCNSB, IIIT Hyderabad <i>Research Assistant</i> - Advised by Prof. Deva Priyakumar, I developed a deep reinforcement learning method, akin to AlphaGo, capable of deducing molecular structures from the Infrared and NMR spectra of the molecule. - Developed a method that predicts structural constraints of RNA using just the RNA sequence as a means to improve the tertiary structure prediction.	Apr. 2021 - July 2024
Chaincode Labs <i>Software Developer</i> - Continuing on work from Summer of Bitcoin, I spearheaded the development of a ‘-nomempool’ runtime option. - This feature allows for the operation of a Bitcoin node without a mempool, allowing users to participate in the Bitcoin network without needing to mine blocks.	Sep. 2021 - Jan 2022
Summer of Bitcoin <i>Student Developer</i> - Streamlined the testing operations of Bitcoin Core by replacing the compiled wallet with a versatile Python mini-wallet. - Additionally, I also reduced the dependency of a Bitcoin node on its mempool, contributing to the system’s overall robustness and performance optimization.	July 2021 - Sep. 2021
Paymatrix <i>Software Development Intern</i> Ported numerous features from the web version of Paymatrix to their mobile application, utilizing Flutter.	Feb. - Apr. 2021
Blue Marble Space Institute of Science <i>Research Associate</i> - As a participant in the Young Scientist Program at BMSIS, I collaborated with Dr. Henderson James Cleaves and Dr. Markus Meringer to benchmark ML-based Electron Ionization Mass Spectrometry (EI-MS) prediction methods. - We also formulated and explored various spectral distance metrics that have not been addressed by other works, to help establish a standard for evaluating ML-based EI-MS prediction methods.	June - Dec. 2020

Summer Research Project

- In collaboration with Dr. Dimitra Atri, I ran simulations using the Geant4 toolkit for a study focused on understanding the impacts of solar radiation on astronauts.
- I completed the Human Phantom Model within the C++ source code of Geant4 and conducted simulations of solar particle events using this model of astronauts.

Relevant Coursework

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| <ul style="list-style-type: none"> • Manycore Parallel Algorithms (UIUC) • Parallel Programming (UIUC) • Compiler Construction (UIUC) • Distributed Systems (UIUC) • Advanced Topics in Computer Architecture (UIUC) • Parallel Computer Architectures (UIUC) • Computer Systems Organization (IIIT-H) • Introduction to Software Systems | <ul style="list-style-type: none"> • (IIIT-H) • Data and Applications (IIIT-H) • Statistical Methods in AI (IIIT-H) • Topics in Deep Learning (IIIT-H) • Advanced NLP (IIIT-H) • ML for Natural Sciences (IIIT-H) • Data Driven Drug Discovery (IIIT-H) • Computer Programming (IIIT-H) • Data Structures and Algorithms (IIIT-H) • Algorithm Analysis and Design | <ul style="list-style-type: none"> • (IIIT-H) • Automata Theory (IIIT-H) • Machine, Data, and Learning (IIIT-H) • Operating Systems and Networks (IIIT-H) • Compilers (IIIT-H) • Design and Analysis of Software Systems (IIIT-H) • Advanced Computer Architecture (IIIT-H) |
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Selected Projects**GPU Spatial Partitioning for Multi-model Workloads****Jan - May 2025**

We evaluated how spatial partitioning on GPUs would impact the working of ML inference pipelines. We found that there is very little overhead due to the additional data movement between spatial partitions, and there is a great advantage in the ML inference latencies being consistent and predictable.

XR Workload Scheduler on a Mobile Device**Jan - May 2025**

We developed multiple schedulers that choose to run ML inference on the CPU/GPU/NPU of a mobile XR device to minimize the overall queuing and inference latency. 2 of our schedulers perform better than using only one of the CPU/GPU/NPU in terms of number of deadlines missed.

Distributed File System**Aug - Dec 2024**

Implemented a replicated, distributed file system in Golang. The filesystem is resilient and adaptable by implementing features such as heartbeat messages to maintain membership and replication, caching to minimize network traffic, and chunking of large files into smaller blocks.

Meta-Stable State Prediction Using Coarse-Grained Models**Feb - April 2023**

Leveraged the latent space of a coarse-grained auto-encoder for predicting the meta-stable states of a molecular system. The approach involved a conditional exploration of the latentent space to identify various stable states of a system.

Racket to x86-64 Compiler**Jan - April 2023**

Implemented a compiler that translates a subset of Racket programming language into x86-64 assembly code. The project emphasized on creating a robust and functional compiler, and understanding the intricacies of compiling to x86-64.

Automation of NMR Spectra Integration Using Machine Learning**June - Sep 2022**

I led a project to understand and automate an existing workflow of a pharmaceutical company. By employing machine learning techniques, we utilized domain-specific knowledge to automate peak identification and integration of NMR spectra to streamline these segments in the company's workflow.

Inexact Machine Unlearning with Graph Neural Networks**Jan - Apr 2022**

Adapted the IC Test pipeline proposed for image classification to graph and node-level tasks. Produced novel results in graph unlearning about random label memorization and proposed unlearning methodologies for GNNs.

LeNet-5 from Scratch**Sep - Nov 2021**

We implemented LeNet-5, a renowned convolutional neural network, entirely from scratch without the aid of any automatic differentiation tools. This project demonstrates a deep understanding of neural network mechanics and the underlying mathematical principles.

Desh: a Custom Linux Shell in C**Oct 2021**

'Desh' is a custom shell designed for Linux-based systems developed from scratch in C, relying exclusively on system calls. This approach required a comprehensive understanding of Linux system architecture and proficient command over C programming, showcasing advanced skills in both system-level programming and operating system concepts.

ASeniorIsAsking**May 2020**

We developed a web application designed to connect students at IIITH with peers at DLF, a popular fast-food hub. The application was built using ReactJS for an intuitive frontend, Redis as the database, and Flask for a robust backend. This project shows an integrated approach to web development, combining various technologies to create a user-friendly platform.

Publications

- Generative artificial intelligence for small molecule drug design** August 2024
Current Opinion in Biotechnology
Ganesh Chandan Kanakala*, **Sriram Devata***, Prathit Chatterjee, Deva Priyakumar
- TorRNA - Improved Prediction of Backbone Torsion Angles of RNA by Leveraging Large Language Models** June 2024
ChemRxiv
Sriram Devata, Deva Priyakumar
- DeepSPInN - Deep reinforcement learning for molecular Structure Prediction from Infrared and ¹³C NMR spectra** March 2024
Digital Discovery, Royal Society of Chemistry
Sriram Devata*, Bhuvanesh S.*, Sarvesh M.*, Yashaswi P. , Siddhartha L., Girish V. , Deva Priyakumar
- Comparative Evaluation of Electron Ionization Mass Spectral Prediction Methods** June 2023
Journal of the American Society for Mass Spectrometry
Sriram Devata, Henderson J. Cleaves, John Dimandja, Christopher A. Heist, Markus Meringer
- Crewed Missions to Mars: Modeling the Impact of Astrophysical Charged Particles on Astronauts and Their Health** Aug 2022
Poster Presentation, American Geophysical Union (AGU) Fall Meeting
Dimitra A., Caitlin M.*, **Sriram Devata***, Konstantin H., Dionysios G., Shireen Mm, Maria V., Giulia C. Bassani, Roberto P., Azza A. Bakr, Tammy W.

Talks

- Compilers, Architecture, and Parallel Computing (CAP) Seminar, UIUC** April 2025
I presented some literature on three major problems and their solutions on serving ML inference pipeline workloads.
- Machine Learning Reading Group, IIITH** Jan - Apr 2021
Within the framework of a paper reading group, I engaged in reading and presenting a series of research papers, each encompassing different domains within Machine Learning. The presentations aimed to foster a deeper understanding of the diverse and evolving landscape of Machine Learning, highlighting its interdisciplinary applications and advancements.
- BlueSciCon II 2020** Aug 2020
I presented a comprehensive talk on NEIMS, one of the state-of-the-art machine learning-based EI-MS prediction methods. The presentation covered the intricate details of NEIMS, its role in the broader context of EI-MS prediction, and its significance in the field of analytical chemistry, demonstrating the potential and impact of machine learning in scientific research and analysis.

Teaching Experience

- Instructor at Mumbai Workshop on Quantum Chemistry (IIT Bombay) June 2024
- Teaching Assistant for CS1.403 Compilers (IIIT-H) Spring 2024
- Teaching Assistant for CS2.201 Computer Systems Organization (IIIT-H) Spring 2023
- Teaching Assistant for CS7.403 Statistical Methods in AI (IIIT-H) Spring 2022
- Teaching Assistant for CS4.301 Data And Applications (IIIT-H) Monsoon 2021
- Teaching Assistant for ML for Chemistry and Drug Design (IHub-Data, IIIT-H) Spring 2022

Technical Skills

Languages: Python, C++, CUDA

Frameworks: PyTorch, RDKit

Web Development: HTML/CSS, JavaScript, Flask, Django, MERN

Utilities/Technologies: Linux, Git, Bash, Slurm, Jupyter, Docker

Awards and Honours

- W.J. Poppelbaum Memorial Award (for graduate students in computer architecture), *UIUC* 2025
- Student Travel Grant, *MLSys'25* 2025
- Qualcomm Innovation Fellowship Finalist 2025
- Student Travel Grant, *ASPLOS'25* 2025
- CS PhD Fellowship, *UIUC* 2024
- Undergraduate Architecture Mentoring Workshop, *ISCA 2023* 2023
- 1st Place Winner at *ML4Science Conference Hackathon, 2023* 2023

Recommended for Poster Session, <i>ACM MICRO 2022 Student Research Competition</i>	2022
ISCA 2022 (Virtual) Student Participation Grant, <i>uArch Workshop</i>	2022
Dean's Merit List for Academic Excellence - Monsoon 2021, <i>IIT Hyderabad</i>	2021
Dean's Merit List for Academic Excellence - Spring 2021, <i>IIT Hyderabad</i>	2021
Dean's Merit List for Academic Excellence - Monsoon 2020, <i>IIT Hyderabad</i>	2020
Dean's Merit List for Academic Excellence - Monsoon 2019, <i>IIT Hyderabad</i>	2019
MEXT Scholarship for Undergraduate Studies (Withheld due to Covid), <i>Government of Japan</i>	2019
UGEE Examination and Interview for CS Research Program - All India Rank 44, <i>IIT Hyderabad</i>	2019

Grants, Resources, and Funding

Understanding Overheads for Multi-Model Workloads in GPU Datacenters	2025
Received resources with an estimated value of \$85,092 through the NSF ACCESS program.	

Professional/Academic Service

Operating Systems Design and Implementation (OSDI'25) Artifact Evaluation Committee	2025
Machine Learning and Systems (MLSys'25) Artifact Evaluation Committee	2025
Judge at UIUC's Undergraduate Research Symposium	2025
Compilers, Architecture, and Parallel Computing (CAP) Seminar Organizing Committee, <i>UIUC</i>	2025-Present

Volunteering/Extracurricular Activities

Member of the Apex Student Council, <i>IITH</i>	2020-2022
Club Coordinator, <i>The Dance Crew, IITH</i>	2020-2022
Collaborations Team Member, Rotating Head, <i>Entrepreneurship Cell, IITH</i>	2019-2022
Head of Corporate Team, <i>Ping! - the student magazine of IITH</i>	2019-2021
Design Team Member, <i>Pentaprism - the photography club of IITH</i>	2019-2020